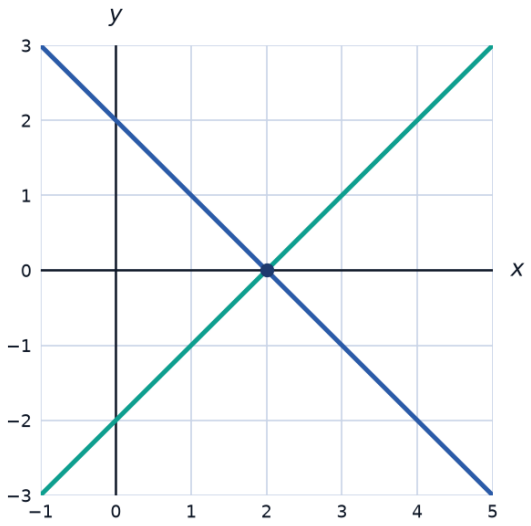
Lines and Planes in Three Dimensions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

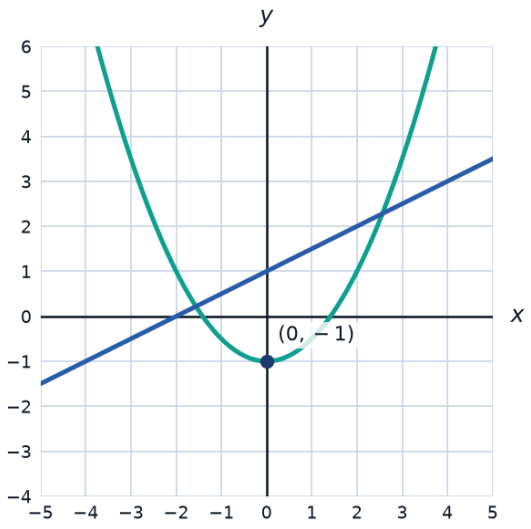
1 [Foundational] • Target: Checking whether a point lies on a plane

This question has three parts. Two lines are given by $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$ and $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$. (a) Show that both lines lie in the plane $z = 1$. (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.



2 [Foundational] • Target: The vector equation of a line

Write the vector equation of the line through the point $(2, -1, 3)$ with direction vector $(1, 2, -1)$.



3 [Foundational] • Target: The vector equation of a line

Find the vector equation of the line through the two points $A(1, 0, 2)$ and $B(4, 3, 1)$.

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4 [Foundational] • Target: Determining if two lines intersect

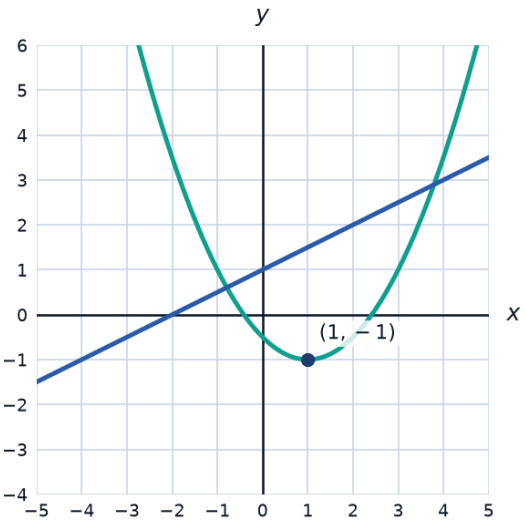
Determine whether the lines $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$ and $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$ intersect, and if so, find the point of intersection.

5 [Foundational] • Target: The equation of a plane

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

6 [Foundational] • Target: The equation of a plane

Find the Cartesian equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, and $C(0, 0, 1)$.



7 [Foundational] • Target: Angle between a line and a plane

Find the angle between the line with direction $(1, 2, 2)$ and the plane $x + y + z = 5$, to the nearest degree.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Distance from a point to a plane

Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.

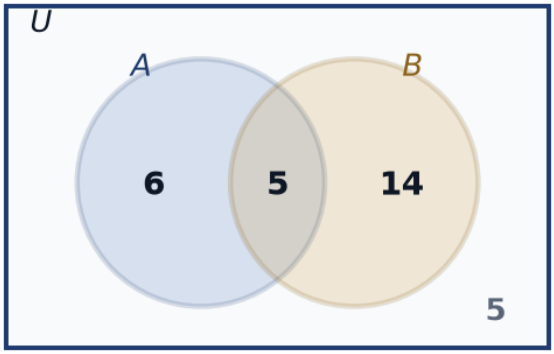
9 [Intermediate] • Target: Finding the angle between two planes

Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

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10 [Intermediate] • Target: Finding the intersection line of two planes

Find a vector equation for the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$, using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.



11 [Intermediate] • Target: Testing whether a line lies within a plane

Determine whether the line $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$ lies entirely within the plane $x + 3y + z = 10$.

12 [Intermediate] • Target: Finding the point where a line meets a plane

Find the point where the line $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$ intersects the plane $2x + y - z = 3$.

13 [Intermediate] • Target: Shortest distance between two skew lines

Two skew lines have direction vectors $\mathbf{d}_1 = (1, 0, 1)$ and $\mathbf{d}_2 = (0, 1, 1)$, and pass through the points $A(0, 0, 0)$ and $B(1, 1, 0)$ respectively. Find the shortest distance between the two lines, using $d = |\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)| / |\mathbf{d}_1 \times \mathbf{d}_2|$.

14 [Intermediate] • Target: Foot of the perpendicular from a point to a line

Find the foot of the perpendicular from the point $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

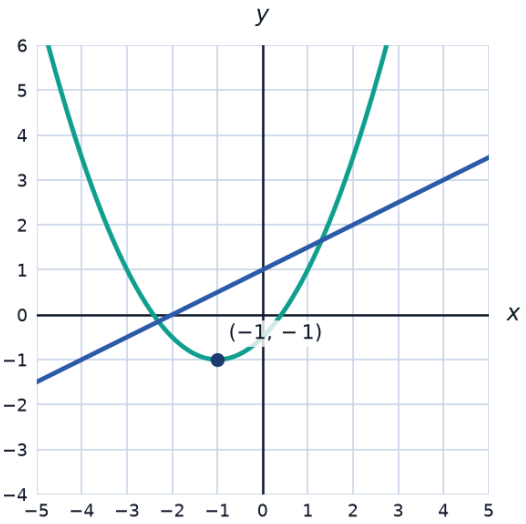
15 [Challenging] • Target: Foot of the perpendicular from a point to a line • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Determine whether the planes $2x - y + 3z = 4$ and $-4x + 2y - 6z = -8$ represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.

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16 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Determine whether the point $P(2,\ -1, 4)$ lies on the plane $3x + 2y - z = 1$, and if not, state whether it lies on the same side as the origin.



17 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Find the value of k such that the line $\mathbf{r} = (1, 2, 0) + t(2,\ -1, k)$ is parallel to the plane $x + 2y - z = 5$.

18 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A surveyor models a hillside as the plane $2x + 3y + 6z = 42$, where x and y are horizontal coordinates (in metres) and z is height (in metres). Find the height z of the hillside at the point where $x = 6$ and $y = 4$, and find the plane's intercept on the z-axis (i.e. the height at $x = 0, y = 0$).

19 [Challenging] • Target: Converting between vector and Cartesian forms of a line • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Write the parametric equations of the line $\mathbf{r} = (2, 1,\ -3) + t(1,\ -2, 4)$, and hence give its Cartesian (symmetric) form.

20 [Challenging] • Target: Determining if two planes are parallel • Cross-Lesson: 4.02 The Poisson Distribution

Determine whether the planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ are parallel, and if so, whether they represent the same plane.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vector and Cartesian Equations of Lines

Find the vector equation of the line passing through point $A(2, -1, 4)$ in the direction of $\mathbf{d} = [3, 5, -2]$.

a. $\mathbf{r} = [2, -1, 4] + \lambda[3, 5, -2]$

b. $\mathbf{r} = [3, 5, -2] + \lambda[2, -1, 4]$

c. $\mathbf{r} = [2 + 3\lambda, -1 - 5\lambda, 4 - 2\lambda]$

d. $\mathbf{r} = [5, 4, 2] + \lambda[1, 1, 1]$

22 [Foundational] • Target: Vector and Cartesian Equations of Lines

Convert the symmetric Cartesian line equations $\frac{x-3}{2} = \frac{y+1}{-4} = \frac{z-5}{1}$ into a direction vector.

a. $[2, -4, 1]$

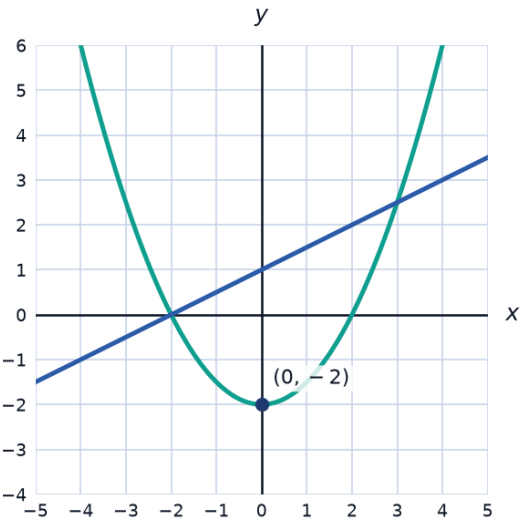
b. $[3, -1, 5]$

c. $[-2, 4, -1]$

d. $[1/2, -1/4, 1]$

23 [Foundational] • Target: Vector and Cartesian Equations of Lines

Two lines in 3D that do not intersect and are not parallel are called:



a. Skew lines

b. Coplanar lines

c. Coincident lines

d. Asymptotic lines

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24 [Foundational] • Target: Vector and Cartesian Equations of Planes

Find the Cartesian equation of the plane passing through $P(1, 2, -3)$ with normal vector $\mathbf{n} = [2, -4, 5]$.

a. $2x - 4y + 5z = -21$

b. $2x - 4y + 5z = 21$

c. $x + 2y - 3z = -21$

d. $2x + 4y + 5z = 0$

25 [Foundational] • Target: Vector and Cartesian Equations of Planes

What is a normal vector to the plane $3x - 7y + z = 14$?

a. $[3, -7, 1]$

b. $[3, 7, 1]$

c. $[14, 14, 14]$

d. $[-3, 7, 14]$

26 [Foundational] • Target: Vector and Cartesian Equations of Planes

Find a vector perpendicular to the plane containing the non-collinear vectors $\mathbf{u}$ and $\mathbf{v}$.

a. $\mathbf{u} \times \mathbf{v}$

b. $\mathbf{u} \cdot \mathbf{v}$

c. $\mathbf{u} + \mathbf{v}$

d. $\mathbf{u} - \mathbf{v}$

27 [Foundational] • Target: Intersections and Angles

Find the intersection point of line $\mathbf{r} = [1 + 2t, -1 + t, 3 - t]$ with plane $x + 2y - z = 4$.

a. $t = 1 \Rightarrow (3, 0, 2)$

b. $t = 0 \Rightarrow (1, -1, 3)$

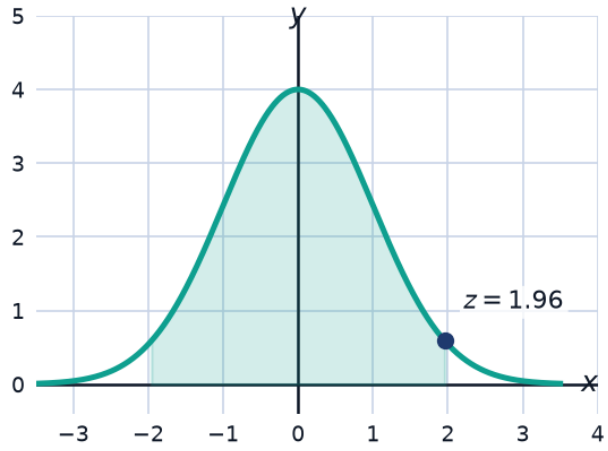
c. $t = 2 \Rightarrow (5, 1, 1)$

d. No intersection

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28 [Foundational] • Target: Intersections and Angles

How is the acute angle θ between a line with direction $\mathbf{d}$ and a plane with normal $\mathbf{n}$ calculated?



a. $\sin\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

b. $\cos\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

c. $\tan\theta = \frac{|\mathbf{d}\times\mathbf{n}|}{|\mathbf{d}\cdot\mathbf{n}|}$

d. $\theta = \arccos(\mathbf{d}\cdot\mathbf{n})$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Intersections and Angles

Find the acute angle between two planes with normal vectors $\mathbf{n}_1 = [1, 0, 1]$ and $\mathbf{n}_2 = [0, 1, 1]$.

a. $\frac{\pi}{3}$ (60°)

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{2}$ (90°)

d. $\frac{\pi}{6}$ (30°)

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30 **[Intermediate]** • **Target: Shortest Distances in 3D**

What is the shortest distance from point $P(x_0, y_0, z_0)$ to the plane $Ax + By + Cz = D$?

a. $\frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$

b. $\frac{|Ax_0 + By_0 + Cz_0 + D|}{A^2 + B^2 + C^2}$

c. $\sqrt{(x_0 - A)^2 + (y_0 - B)^2 + (z_0 - C)^2}$

d. $\frac{|D|}{\sqrt{A^2 + B^2 + C^2}}$

31 **[Intermediate]** • **Target: Shortest Distances in 3D**

Find the distance from the origin $(0, 0, 0)$ to the plane $2x - 3y + 6z = 28$.

a. 4

b. 28

c. $\frac{28}{7} = 4$

d. $\frac{28}{49}$

32 **[Intermediate]** • **Target: Shortest Distances in 3D**

How is the shortest distance between two parallel planes $2x + y - 2z = 5$ and $2x + y - 2z = 14$ calculated?

a. $\frac{|14 - 5|}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{9}{3} = 3$

b. $\frac{14 + 5}{3} = \frac{19}{3}$

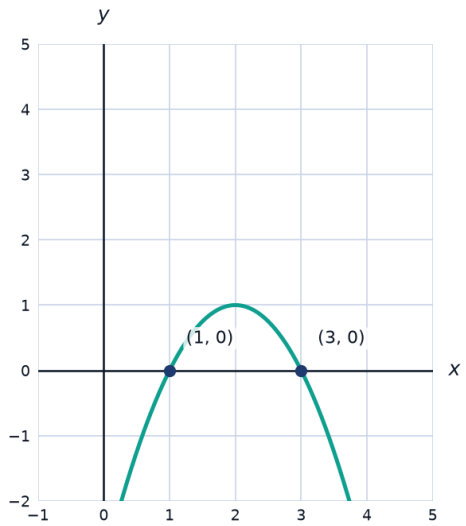
c. $14 - 5 = 9$

d. $\frac{9}{9} = 1$

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33 [Intermediate] • Target: Converting between vector and Cartesian forms of a line

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?



a. 1

b. 2

c. 3

d. 4

34 [Intermediate] • Target: Determining if two planes are parallel

A line has direction $\mathbf{d} = (2; -1; 0)$ and a plane has normal $\mathbf{n} = (1; 2; 2)$. Find the angle between the line and the plane.

a. 30°

b. 45°

c. 0°

d. 90°

35 [Intermediate] • Target: Distance between two parallel planes

A line has direction $\mathbf{d} = (2; -6; 0)$ and a plane has normal $\mathbf{n} = (6; 2; 3)$. Find the angle between the line and the plane.

a. 90°

b. 45°

c. 30°

d. 0°

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36 [Intermediate] • Target: Checking whether three points are collinear

A line has direction $\mathbf{d} = (4;\ -1;\ 0)$ and a plane has normal $\mathbf{n} = (1; 4; 8)$. Find the angle between the line and the plane.

a. 0°

b. 45°

c. 90°

d. 30°

37 [Intermediate] • Target: Reflecting a point in a plane

A line has direction $\mathbf{d} = (3;\ -2;\ 0)$ and a plane has normal $\mathbf{n} = (2; 3; 6)$. Find the angle between the line and the plane.

a. 0°

b. 45°

c. 30°

d. 90°

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Vector and Cartesian Equations of Lines • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A line has direction $\mathbf{d} = (4;\ -4;\ 0)$ and a plane has normal $\mathbf{n} = (4; 4; 7)$. Find the angle between the line and the plane.

a. 0°

b. 45°

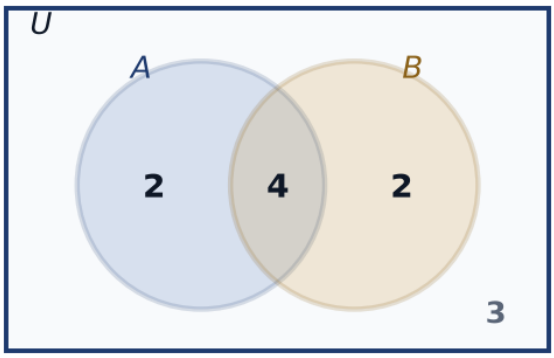
c. 90°

d. 30°

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39 [Challenging] • Target: Vector and Cartesian Equations of Planes • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Line: $\mathbf{r} = (2; 4; -1) + t(2; 1; -3)$. Plane: $-2x - 2y + 2z = -26$. Find the value of t at the point of intersection.



a. 1

b. -1

c. 4

d. 2

40 [Challenging] • Target: Intersections and Angles • Cross-Lesson: 4.02 The Poisson Distribution

Line: $\mathbf{r} = (1; -3; -2) + t(1; -3; 0)$. Plane: $2x - 3y + 4z = -19$. Find the value of t at the point of intersection.

a. 0

b. 1

c. -2

d. -4

41 [Challenging] • Target: Shortest Distances in 3D • Cross-Lesson: 5.01 Further Integration Techniques

Line: $\mathbf{r} = (2; -3; 0) + t(-3; 0; 2)$. Plane: $-x - 3y - 3z = 16$. Find the value of t at the point of intersection.

a. -4

b. -5

c. 0

d. -3

42 [Challenging] • Target: The vector equation of a line • Cross-Lesson: 5.02 Differential Equations

Line: $\mathbf{r} = (1; 5; 4) + t(2; 0; -1)$. Plane: $4x - 2y = 10$. Find the value of t at the point of intersection.

a. 4

b. 2

c. 0

d. -1

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43 [Challenging] • Target: Determining if two lines intersect • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Line: $\mathbf{r} = (0;\ -1;\ -5) + t(0;2;\ -2)$. Plane: $4x - 3z = 27$. Find the value of t at the point of intersection.

a. 2

b. -1

c. 0

d. 3

44 [Challenging] • Target: The equation of a plane • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Line: $\mathbf{r} = (4;\ -2;0) + t(3;\ -2;1)$. Plane: $3x + y - z = 16$. Find the value of t at the point of intersection.

a. 1

b. 2

c. -2

d. 3

45 [Challenging] • Target: Angle between a line and a plane • Cross-Lesson: 6.01 Matrices (AI HL)

Line: $\mathbf{r} = (-3;\ -4;\ -1) + t(3;0;2)$. Plane: $2x - 2y - z = -1$. Find the value of t at the point of intersection.

a. 1

b. -3

c. -1

d. -4

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